

2 INPhO-2007

1. (a)

$$\text{Given } r = at, \theta = \omega t \quad (29)$$

$$\text{Now } \vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$

$$\text{Hence } \vec{p} = m(a\hat{r} + r\omega\hat{\theta})$$

$$\vec{F} = m\vec{a}$$

$$= m[(\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}]$$

$$= m(-r\omega^2\hat{r} + 2a\omega\hat{\theta})$$

(b)

$$\int F_r dr = -m\omega^2 \int_0^r r dr = \frac{-m\omega^2 r^2}{2} \quad (30)$$

From Eq. (29)

$$\theta = \frac{r\omega}{a}$$

$$\int F_\theta r d\theta = 2m\omega^2 \frac{r^2}{2} = m\omega^2 r^2 \quad (31)$$

Adding Eqs. (30) and (31)

$$\Delta W = \frac{m\omega^2 r^2}{2}$$

We can also obtain the above result by employing the work-energy theorem.

(c) Trajectory will be a spiral.

2. (a) From Fig. (11)

$$v_y = v \sin \theta \quad (32)$$

Note that the surface is rough and there is frictional force along the x -direction. Hence elastic collision does not constrain the velocity along the x -direction. It implies that the y -component of the velocity $v \sin \theta$ changes only in sign. From Newton's second law along vertical y -direction, change in momentum is given by the linear impulse, which yields:

$$2mv \sin \theta = \int N dt \quad (33)$$

From Newton's second law along x -direction, we have

$$mv \cos \theta - \mu \int N dt = mv_x \quad (34)$$

Inserting Eq. (33) in Eq. (34) we obtain

$$v_x = v(\cos(\theta) - 2\mu \sin(\theta))$$

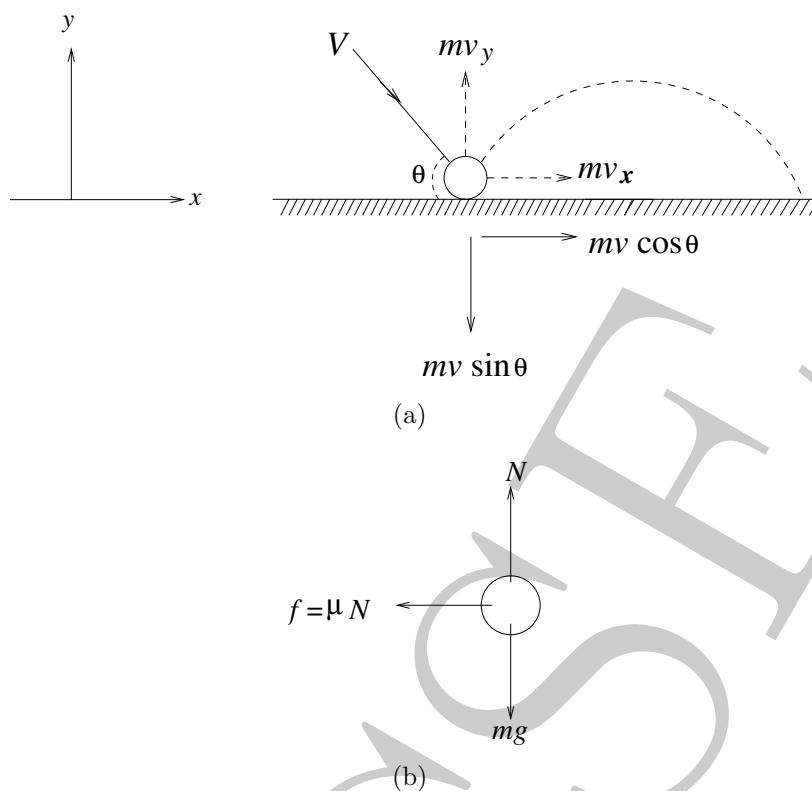


Figure 11: (a) Problem 2. (b) Free body diagram of the ball in contact with the floor.

$$\text{Range} = v_x \times \text{time of flight} = v_x \times \frac{2v_y}{g}$$

$$R(\theta) = \frac{2v^2}{g} f(\theta)$$

where

$$f(\theta) = \sin \theta (\cos \theta - 2\mu \sin \theta)$$

To maximize R , set

$$f'(\theta_m) = 0$$

which yields

$$\theta_m = \frac{1}{2} \cot^{-1}(2\mu)$$

(b) Possible range of θ_m :

$$\theta_m \in]0, \pi/4[$$

3. See Fig. (12).

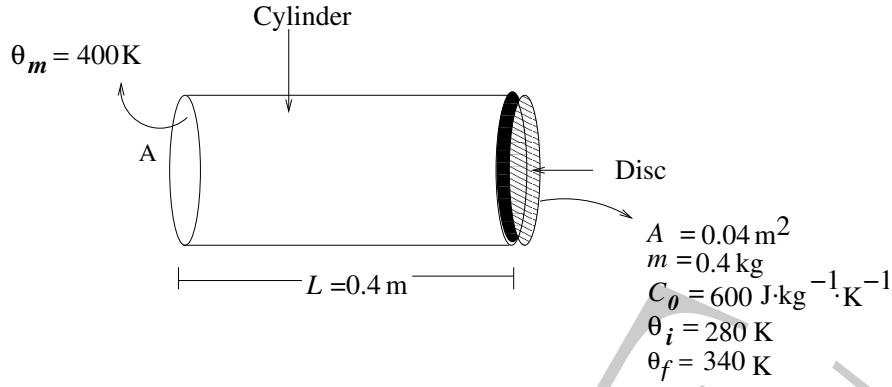


Figure 12: Problem 3

(a) By Fourier's law of heat conduction

$$\frac{dQ}{dt} = \frac{KA(\theta_m - \theta)}{L} \quad (35)$$

By calorimetry

$$\frac{dQ}{dt} = mC(\theta) \frac{d\theta}{dt} \quad (36)$$

By Eqs. (35) and (36)

$$\frac{1 - \alpha(\theta_i - \theta_m)}{\theta_m - \theta} d\theta - \alpha d\theta = \frac{KA}{mC_0 L} dt$$

Integrating and noting that $mLC_0/KA = 240\text{ s}$

$$(1 - \alpha(\theta_i - \theta_m)) \ln \left(\frac{\theta_m - \theta_i}{\theta_m - \theta_f} \right) - \alpha(\theta_f - \theta_i) = \frac{t}{240} \quad (37)$$

Inserting the values $t \approx 222\text{ s}$.

(b) If $\alpha = 0$, then from Eq. (37)

$$\begin{aligned} \frac{t}{240} &= \ln \left(\frac{\theta_m - \theta_i}{\theta_m - \theta_f} \right) \\ t &= 166\text{ s} \end{aligned}$$

(c) Process (3a) takes more time since as the disk heats up, its specific heat also increases and more heat is required to effect a further rise in temperature. Note

$$C(\theta_i) = 600\text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$$

$$C(\theta_f) = 960\text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$$

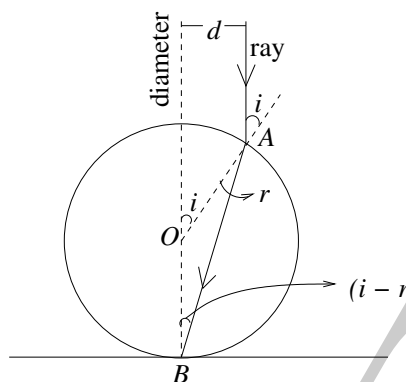


Figure 13: Problem 4

- (a) From sine rule for $\triangle OAB$ (see Fig. (13))

$$\frac{OA}{\sin(i-r)} = \frac{OB}{\sin r}$$

$$\begin{aligned} OA = OB &= R \\ \therefore \sin r &= \sin(i-r) \\ &= \sin i \cos r - \cos i \sin r \end{aligned}$$

Using Snell's law for refraction

$$\text{Hence } \frac{1}{n} = \frac{\sin i}{\sqrt{n^2 - \sin^2 i}} = \frac{n \sin r}{\sqrt{1 - \sin^2 i}}$$

From Fig. (13), $\sin i = d/R$.
Solving above equations

$$d^2 = R^2 \left(1 - \frac{(n^2 - 2)^2}{4} \right)$$

- (b) d ranges from 0 to R .

If $d \rightarrow 0$.

$$n = \pm 2 \text{ or } 0$$

Only physically allowed value is $n = +2$.

If $d \rightarrow R$.

$$n = \pm\sqrt{2}$$

Only allowed value is $+\sqrt{2}$.

Hence the allowed range of n is

$$n \in]\sqrt{2}, 2[$$

5. (a) Using Gauss law, the electric field is

$$E(r) = \frac{\lambda}{2\pi\epsilon_0 r}$$

- (b) Integrating along a radial line, we get the electric potential

$$V(r) = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r}{a}\right)$$

where a is the distance of reference point from the line charge.

- (c) Total potential at point $P \equiv (x, y)$ is (see Fig. (14))

$$V(P) = -\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_+}{r_-}\right)$$

Equipotential lines in $z = 0$ plane are given by the equation $V(P) = V_0$, some constant. Let $k = \exp(2\pi\epsilon_0 V_0/\lambda)$. Then equation for equipotential lines becomes

$$\begin{aligned} \frac{r_-}{r_+} &= k \\ \frac{(x+d)^2 + y^2}{(x-d)^2 + y^2} &= k^2 \\ \left(x - d\frac{k^2+1}{k^2-1}\right)^2 + y^2 &= \frac{4d^2k^2}{(k^2-1)^2} \end{aligned}$$

This is an equation of a circle with centre at $\left(d\frac{k^2+1}{k^2-1}, 0\right)$ and radius

given by $\frac{2dk}{|k^2-1|}$.

- (d) See Fig. (14)

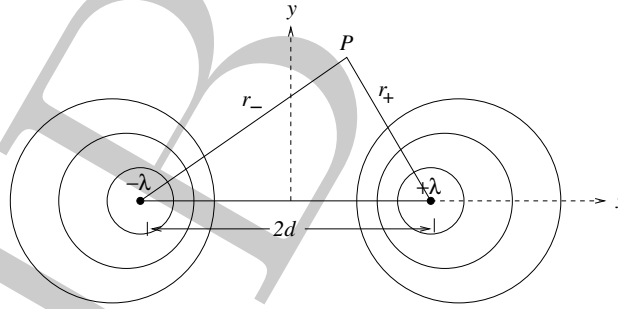


Figure 14: Problem 5 (d)

- (e) Magnitude of electric force per unit length is given by

$$F_E = \frac{\lambda^2}{4\pi\epsilon_0 d}$$

Magnitude of magnetic force per unit length is given by

$$F_M = \frac{\mu_0 \lambda^2 v^2}{4\pi d}$$

If these are equal then $v = 1/\sqrt{\epsilon_0\mu_0} = c$. This emphasizes the general observation that magnetic force is very small compared to the electrostatic force.

6.

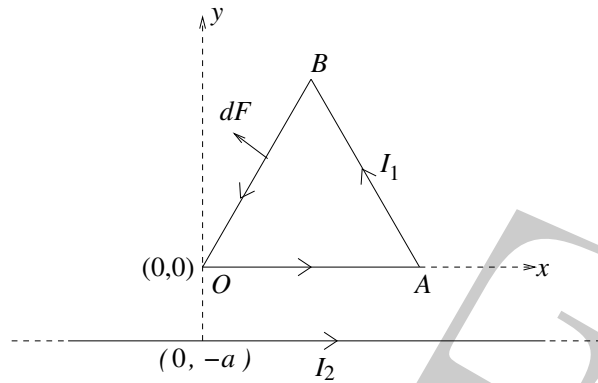


Figure 15: Problem 6 (a)

(a) See Fig. (15). Force F_1 on side OA is

$$F_1 = \frac{\mu_0 I_2 I_1 S}{2\pi a}$$

downwards, i.e. the negative y - direction.

Consider element dr at a distance r from O on OB . On both of the sides (OB and AB) same magnitude of force acts. The x components cancel and the y components add. Hence total force on both sides

$$F_{2y} = \frac{\mu_0 I_1 I_2}{2\pi} \int_0^S \frac{1}{a + \sqrt{3}r/2} dr$$

upwards, i.e. the positive y - direction.

By adding these forces, total force on the triangle is

$$F = \frac{\mu_0 I_1 I_2}{\sqrt{3}\pi} \left[\frac{\sqrt{3}S}{2a} - \ln \left(1 + \frac{\sqrt{3}S}{2a} \right) \right]$$

This net force will be downwards.

(b) See Fig. (16). The behaviour is quadratic for small S/a and almost linear for large S/a .

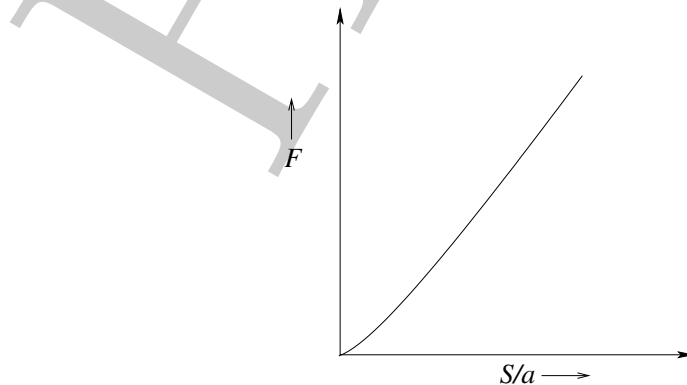


Figure 16: Problem 6 (b)

7. (a) Let n be the number of photons per second and P , the power of the source.

Change in momentum of one photon

$$= \frac{h}{\lambda}(\hat{j} - \hat{i})$$

Rate of change of linear momentum of the beam is

$$\begin{aligned}\vec{F} &= \frac{nh}{\lambda}(\hat{j} - \hat{i}) \\ &= \frac{P}{c}(\hat{j} - \hat{i}) \quad \left(\because P = \frac{nhc}{\lambda}\right)\end{aligned}$$

Inserting values

$$|\vec{F}| = 2.82 \times 10^{-7} \text{ N}$$

- (b) Surface charge density σ of metal surface after time t with photoelectric efficiency η .

$$\begin{aligned}&= \frac{n}{A} e t \eta \\ &= \frac{P \lambda}{hcA} e t \eta\end{aligned}$$

Now $t = 10$ s and $\eta = 0.1$. Hence

$$\sigma = 7.27 \times 10^2 \text{ C}\cdot\text{m}^{-2}$$

- (c) Energy density after 10 s

$$\begin{aligned}&= \frac{1}{2} \epsilon_0 \left(\frac{\sigma}{\epsilon_0} \right)^2 \\ &= 2.93 \times 10^{16} \text{ J}\cdot\text{m}^{-3}\end{aligned}$$

- (d) Energy of one photon

$$E = \frac{hc}{\lambda} = 2.06 \text{ eV}$$

Given that work function

$$\phi = 1.9 \text{ eV}$$

Then

$$K_{\max} = E - \phi = 0.16 \text{ eV}$$

Hence the range of kinetic energy is from 0 to 0.16 eV.

8.

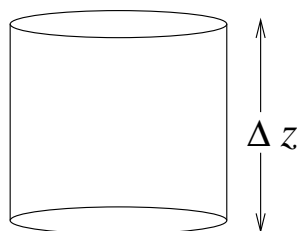


Figure 17: Problem 8

- (a) From Fig. (17)

$$\Delta p = -\rho g \Delta z$$

Let m_a be the mass of one mole of air. Then by ideal gas law

$$p = \frac{\rho RT}{m_a}$$

Also, the equation for adiabatic process is $p^{\gamma-1} \propto T^\gamma$

From these equations,

$$\Delta T = -\frac{(\gamma-1)}{\gamma} \frac{m_a g}{R} \Delta z$$

The lapse rate equation is

$$\Gamma = -\frac{(\gamma-1)}{\gamma} \frac{m_a g}{R}$$

- (b) For diatomic gas $\gamma = 7/5$ and $m_a = 2.9 \times 10^{-2} \text{ kg} \cdot \text{mol}^{-1}$.
Hence $\Gamma = -0.01 \text{ K} \cdot \text{m}^{-1}$ (i.e. 1°C decrease for every 100 m).
(c) Using equations of part (8a)

$$\frac{dp}{p} = -\frac{m_a g}{R} \frac{dz}{(T_0 - \Gamma z)}$$

On integrating

$$p = p_0 \left(\frac{T_0 - \Gamma z}{T_0} \right)^{m_a g / R \Gamma}$$

There are two interesting aspects about the exponent of the above equation. The first is a rare occurrence of “Force” related dimension in gases, namely $R\Gamma$. The second is its numerical value $m_a g / R\Gamma \simeq 3$.

For reasonable heights (e.g. $z = 1 \text{ km}$)

$$p(z) \simeq p_0 \left[1 - \frac{m_a g z}{RT_0} \right]$$

- (d) Inserting given values in the above equation, the height of the atmosphere is approximately 30 km.
9. (a) The magnetic field at the centre of the field coil is

$$B = \frac{\mu_0}{2} \frac{N_f}{R_f} I(t)$$

- (b) Flux on the pick-up coil

$$\phi = \pi R_p^2 B$$

Induced emf in N_p turns

$$\begin{aligned} \varepsilon &= -N_p \frac{d\phi}{dt} \\ |\varepsilon| &= \frac{\mu_0}{2} \frac{N_p N_f}{R_f} \pi R_p^2 \frac{dI(t)}{dt} \end{aligned}$$

(c) For maximum ε

$$\varepsilon_0 = \frac{\mu_0}{2} \frac{N_p N_f}{R_f} \pi R_p^2 I_0 \omega$$

$$\therefore N_p N_f = 645 \quad (38)$$

(d) Mutual inductance on field coil due to pick up coil is equal to mutual inductance on pick up coil due to field coil. Hence mutual inductance

$$M = \frac{\mu_0}{2} \frac{N_p N_f}{R_f} \pi R_p^2 = 1.59 \times 10^{-5} \text{ H}$$

(e) The length of wire used is

$$L = 2\pi R_f N_f + 2\pi R_p N_p$$

To optimize it

$$\frac{dL}{dN_f} = 0$$

Eq. (38) yields

$$N_f = 18 \text{ turns}, N_p = 36 \text{ turns}$$

(f) Induced emf in case of

- i. Iron : will increase.
- ii. Wood : no appreciable change.
- iii. Copper: decrease.



Max Planck (23 Apr.1858-4 Oct.1947) : Max (Karl Ernst Ludwig) Planck was a German theoretical physicist. He studied at Munich and Berlin, where he studied under Helmholtz, Clausius and Kirchoff and subsequently joined the faculty. He became professor of theoretical physics (1889-1926). His work on the law of thermodynamics and the distribution of radiation from a black body led him to abandon classical Newtonian principles and introduce the quantum theory (1900). For this he was awarded the Nobel Prize for Physics in 1918.

This assumes that energy is not infinitely divisible, but ultimately exists as discrete amounts he called quanta (Latin, “how much”). Further, the energy carried by a quantum depends in direct proportion to the frequency of its source radiation.

The work leading to the “lucky” blackbody radiation formula was described by Planck in his Nobel Prize acceptance speech (1920): **“But even if the radiation formula proved to be perfectly correct, it would after all have been only an interpolation formula found by lucky guess-work and thus, would have left us rather unsatisfied. I therefore strived from the day of its discovery, to give it a real physical interpretation and this led me to consider the relations between entropy and probability according to Boltzmann’s ideas. After some weeks of the most intense work of my life, light began to appear to me and unexpected views revealed themselves in the distance.”** [See also the Foreword]